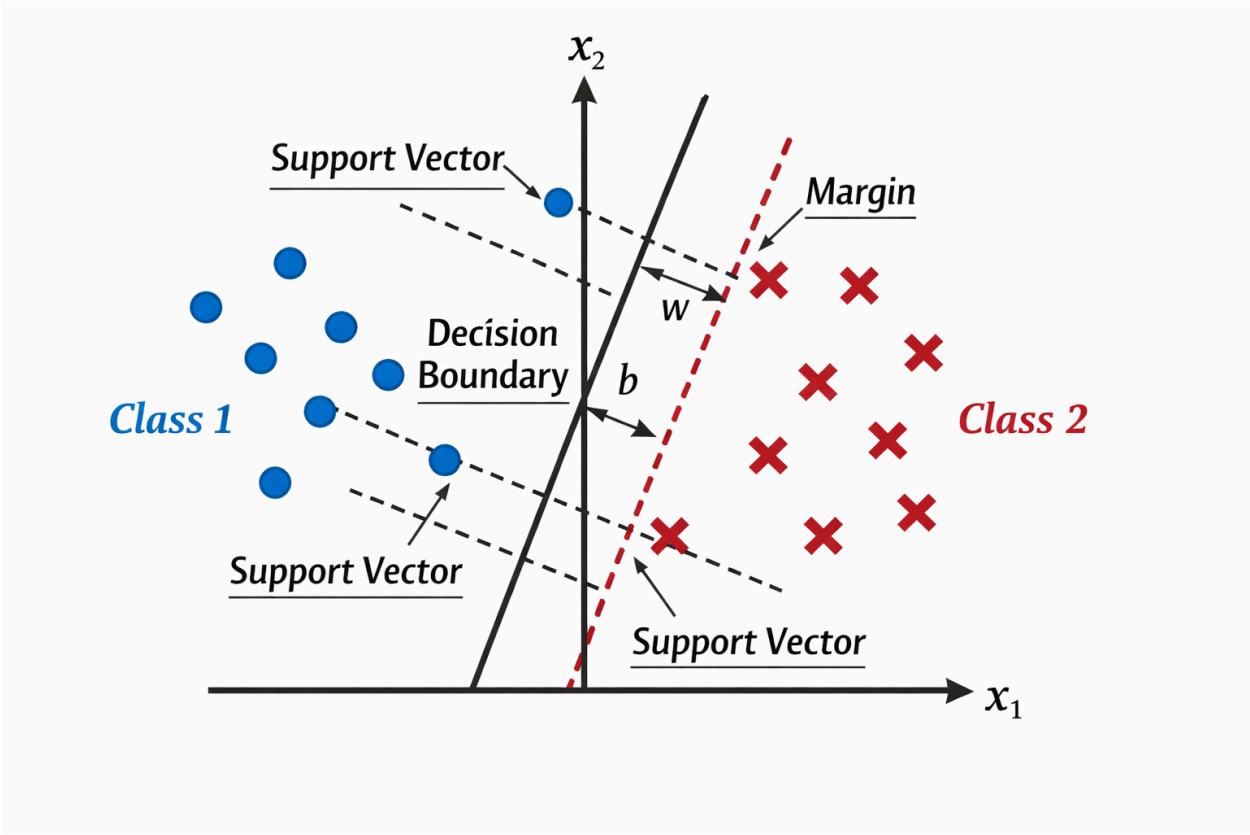
Documentation-Support Vector Machine (SVM)



A **Support Vector Machine (SVM)** is a supervised learning algorithm used for classification and regression.

The main idea: Find the **optimal hyperplane** that separates classes with the **maximum margin**.

Key Components of SVM:

1 Hyperplane

A decision boundary: $w^T x + b = 0$

2 Margin

Distance between hyperplane and closest points.

Margin = 2 / ||w||

3 Support Vectors

Training points that lie closest to the hyperplane.
They determine the decision boundary.

4 Hard Margin vs Soft Margin

- **Hard Margin** → Data perfectly separable
- **Soft Margin** → Allows misclassification using slack variables

5 Kernel Trick

Maps data into higher dimensional space: $K(x_i, x\square) = \varphi(x_i)^T \varphi(x\square)$

Common kernels:

- Linear
- Polynomial
- RBF (Gaussian)
- Sigmoid

1 Problem Formulation

1.1 Input Space

$X \in \mathbb{R}^d$

Training dataset: $\{(x_i, y_i)\}$ for $i = 1,\ldots,n$

Where: $x_i \in \mathbb{R}^d$, $y_i \in \{-1, +1\}$

1.2 Output Space

$y \in \{-1, +1\}$

1.3 Data Distribution

$(x, y) \sim P(X, Y)$

Unknown joint distribution.

1.4 Learning Objective (Expected Risk Minimization)

Minimize expected 0–1 loss: $R(f) = \mathbb{E}[\, I(y \neq f(x)) \,]$

Since 0–1 loss is non-convex, SVM uses hinge loss surrogate.

2 Model Specification

2.1 Hypothesis Function

$f(x) = w^T x + b$

Decision rule: $\hat{y} = \text{sign}(w^T x + b)$

2.1 Parameter Space

$\mathbf{w} \in \mathbb{R}^d$

$\mathbf{b} \in \mathbb{R}$

2.2 Structural Assumptions

- Linear separability (hard margin case)
- Maximum margin principle
- Kernel assumption (if nonlinear)
- Convex optimization framework

3 Loss Function

3.1 Explicit Mathematical Form

Hinge loss: $\ell(y, f(x)) = \max(0, 1 - y(w^T x + b))$

3.2 Why This Loss? (Statistical Reasoning)

- Convex upper bound to 0–1 loss
- Encourages large margin
- Penalizes only misclassified or margin-violating points

3.3 Convexity Properties

Hinge loss is convex but not differentiable at $1 - y(w^T x + b) = 0$.

Thus global minimum exists.

4 Objective Function

4.1 Empirical Risk Expression

Hard Margin SVM

Minimize: $(1/2)\,||w||^2$

Subject to: $y_i(w^T x_i + b) \geq 1$

Soft Margin SVM

Introduce slack variables $\xi_i \geq 0$

Minimize: $(1/2)\ ||w||^2 + C \sum \xi_i$

Subject to: $y_i(w^T x_i + b) \geq 1 - \xi_i$

Empirical Risk Form

Minimize: $(1/2)\ ||w||^2 + C \sum \max(0, 1 - y_i(w^T x_i + b))$

5 Optimization Method

5.1 Closed-form or iterative?

❌ No closed-form solution
✔ Solved via Quadratic Programming (QP)

Dual Formulation

Maximize:

$\sum \alpha_i - (1/2) \sum\sum \alpha_i\ \alpha_{\square}\ y_i\ y_{\square}\ x_i^T x_{\square}$

Subject to:

$0 \leq \alpha_i \leq C$
$\sum \alpha_i\ y_i = 0$

Kernel Version

Replace $x_i^T x_{\square}$ with:

$K(x_i, x_{\square})$

5.2 Gradient expression(Subgradient for Hinge Loss)

If $y_i(w^T x_i + b) \geq 1$:

$\nabla w = w$

Else: $\nabla w = w - C\ y_i\ x_i$

5.3 Convergence guarantees

- Convex optimization
- Global optimum guaranteed
- SMO algorithm commonly used

5.4 Computational complexity

Training:

- $O(n^3)$ worst case (QP solver)
- SMO practical complexity $\approx O(n^2)$

6 Statistical Interpretation

6.1 MLE / MAP connection(Regularization View)

Minimizing: $(1/2)\ \|w\|^2$

Equivalent to:

Gaussian prior on w:

$w \sim N(0, I)$

6.2 Noise model assumption

Loss View: Hinge loss approximates 0–1 loss.

6.3 Probabilistic meaning of outputs

SVM is not inherently probabilistic.
Probabilities obtained via: Platt scaling.

7 Regularization & Generalization

7.1 Bias–variance tradeoff

Parameter C controls tradeoff:

- Large C → low bias, high variance
- Small C → high bias, low variance

7.2 Overfitting behavior

Less prone to overfitting because:

- Maximizes margin
- Depends only on support vectors

7.3 Capacity control mechanism

Controlled by margin size:

VC-dimension bounded by: $\min(\|w\|^2, R^2)$

Larger margin → better generalization.

8 Theoretical Properties

8.1 Convexity / Global optimality

Optimization problem is convex.

Global optimum guaranteed.

8.2 Consistency (if applicable)

Under proper regularization:

SVM is statistically consistent.

8.3 Stability considerations

Relatively stable due to margin maximization.

9 Computational Complexity

9.1 Training time complexity

- Linear SVM (SGD): $O(n\ d)$
- Kernel SVM: $O(n^2$ to $n^3)$

9.2 Inference time complexity

Linear SVM: $O(d)$

Kernel SVM: $O(\#\text{support vectors} \times d)$

9.3 Memory complexity

Linear: $O(d)$

Kernel: $O(n^2)$ (for kernel matrix)

10 Limitations

10.1 When It Fails

- Performs poorly when classes heavily overlap.
- Struggles with very large datasets (especially kernel SVM).
- Not suitable when number of samples n is much larger than features and computation is limited.
- Fails to capture complex structure if wrong kernel is chosen.
- Not ideal when probabilistic outputs are required directly.

10.2 Assumption violations

- Assumes margin-based separability (soft or hard).
- Assumes chosen kernel correctly represents feature similarity.
- Assumes data is reasonably clean and well-labeled.
- Linear SVM assumes linear decision boundary.
- Performance degrades if hyperparameter C is poorly tuned.

10.3 Sensitivity Issues

- Sensitive to feature scaling (requires normalization/standardization).
- Sensitive to outliers (especially with large C).
- Kernel methods sensitive to kernel parameters (e.g., γ in RBF).
- High memory usage in kernel SVM (stores support vectors).
- Training time increases rapidly with dataset size